
Preprocessing Strategies, Regularisation, and Generalisation in Epileptic Seizure Prediction: A Logistic Regression Study

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Abstract

This paper investigates how preprocessing pipeline design, regularisation strategy, and class imbalance handling affect the generalisation performance of logistic regression models in epileptic seizure prediction. Three publicly available EEG-based datasets are used: the UCI Epileptic Seizure Recognition dataset, the BEED Bangalore EEG Epilepsy dataset, and the CHB-MIT Clinical EEG dataset. Two preprocessing pipelines are compared — one emphasising noise removal and feature selection (Pipeline A), and one emphasising feature extraction and dimensionality reduction via PCA (Pipeline B). L1, L2, and Elastic Net regularisation are evaluated across a range of penalty strengths. Class imbalance is addressed via SMOTE, random undersampling, and class weighting. Results show that preprocessing order has the largest single impact on F1-score (mean absolute difference of 0.378), that L2 regularisation generalises most stably across datasets, and that Elastic Net does not consistently outperform L1 or L2 in this setting. These findings highlight the primacy of preprocessing decisions over model hyperparameter tuning in low-complexity classifiers.

Index Terms — epileptic seizure prediction, logistic regression, regularisation, preprocessing pipeline, class imbalance, EEG classification

I. Introduction

Epilepsy affects approximately 50 million people worldwide [1], and automated seizure detection from EEG signals remains a critical open problem in clinical neuroscience. While deep learning approaches have gained traction in recent years, logistic regression remains a relevant and interpretable baseline, particularly given the high dimensionality and class imbalance characteristics of clinical EEG data.

This study examines a deliberately constrained setting: logistic regression with varying preprocessing pipelines, regularisation strategies, and imbalance-handling techniques. The goal is not to achieve state-of-the-art performance, but to systematically isolate the contribution of each design decision to generalisation.

Four research questions guide the analysis:

1. Does preprocessing order (pipeline design) significantly affect classification performance?
 2. Which regularisation type generalises best across heterogeneous datasets?
 3. Does Elastic Net consistently outperform L1 and L2?
 4. How does imbalance handling interact with regularisation choice?
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II. Datasets

Three datasets were selected to ensure diversity in size, imbalance ratio, feature type, and clinical context.

TABLE I — Dataset Summary

ID	Dataset	Source	Samples	Features	Feature Type	Imbalance	Size
D1	UCI Epileptic Seizure Recognition	Bonn University / GitHub	11,500	178	178 EEG time-points per row	4:1 (non-seizure heavy)	~17 MB
D2	BEED Bangalore EEG Epilepsy	Neurological Centre, Bangalore	8,000	16	16 raw scalp channel readings	1:3 (seizure heavy)	~1 MB
D3	CHB-MIT Clinical EEG	Boston Children's Hospital	3,546	5,888	23 channels $\times$ 256 time-points	~1:1 (balanced)	~167 MB

D1 (UCI) contains 11,500 samples of 178 EEG voltage measurements per row, binarised from a five-class label (class 1 = seizure, classes 2–5 = non-seizure), yielding a 4:1 imbalance ratio. **D2 (BEED)** provides 16 raw scalp channel readings across 8,000 samples, originally four-class, binarised to healthy vs. any seizure state, with an inverted imbalance (seizures are the majority class at 3:1). **D3 (CHB-MIT)** is derived from raw 23-channel clinical EEG segments of 256 time-points, reshaped to a 2D feature matrix of 5,888 columns, and is approximately balanced (1:1).

Train/validation/test splits were 70/15/15:

- D1 (df1): train 8,050 / val 1,725 / test 1,725
- D2 (df2): train 5,600 / val 1,200 / test 1,200
- D3 (df3): train 2,482 / val 532 / test 532

III. Preprocessing Pipelines

A key hypothesis of this study is that the ordering of preprocessing steps materially affects performance. Two pipelines were designed with deliberately different philosophies.

Pipeline A is a signal-cleaning-first approach:

- 1. Min-Max Normalisation — scale all features to [0, 1]
- 2. Noise Removal — band-pass filtering / smoothing applied to raw features
- 3. Feature Selection — univariate statistical selection retaining the top-k features

Pipeline B is a representation-learning approach:

- 1. Feature Extraction — statistical descriptors (mean, variance, skewness, kurtosis, zero-crossing rate) extracted per channel or segment
- 2. Standard Scaling — z-score standardisation
- 3. PCA — dimensionality reduction retaining components explaining the target variance threshold

TABLE II — Pipeline Output Dimensionality

Dataset	Pipeline A Features	Pipeline B Components	PCA Variance Explained
df1	50	50	82.88%
df2	10	10	92.92%
df3	100	100	67.69%

The rationale for comparing these two pipelines is that Pipeline A operates on raw signal structure (preserving temporal locality), while Pipeline B transforms the representation space via statistical summarisation followed by orthogonal decomposition. The ordering difference — whether domain knowledge or data-driven reduction comes first — is expected to have downstream consequences on model behaviour.

IV. Baseline Model

Logistic regression was used as the core classifier:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}}$$

The model was trained with L2 regularisation at C=1.0 as the default baseline. Three metrics are reported: **Accuracy**, **F1-score** (macro-weighted), and **PR-AUC** (Precision-Recall Area Under Curve — the primary metric given class imbalance).

TABLE III — Baseline Results (Validation Split)

Pipeline	Dataset	Accuracy	F1-Score	PR-AUC
A	df1	0.805	0.051	0.439
A	df2	0.807	0.886	0.740
A	df3	0.526	0.452	0.547
B	df1	0.946	0.856	0.948
B	df2	0.976	0.984	0.999
B	df3	0.712	0.683	0.804

TABLE IV — Baseline Results (Test Split)

Pipeline	Dataset	Accuracy	F1-Score	PR-AUC
A	df1	0.803	0.034	0.465

A	df2	0.812	0.888	0.734
A	df3	0.491	0.399	0.517
B	df1	0.943	0.845	0.934
B	df2	0.976	0.984	0.998
B	df3	0.754	0.727	0.847

Pipeline B consistently outperforms Pipeline A across all three datasets. The most striking gap occurs on df1, where Pipeline A achieves an F1-score of only 0.051 compared to 0.856 under Pipeline B — a difference of 0.805 F1 points. This is examined in Section VII.

V. Overfitting and Underfitting

To demonstrate the bias-variance tradeoff, two deliberate extremes were constructed on df1.

Underfitting scenario: $C = 0.001$ (strong regularisation), only 3 features used.

- Validation: Acc 0.800, F1 0.000, PR-AUC 0.447
- Test: Acc 0.800, F1 0.000, PR-AUC 0.426

The model collapses to predicting the majority class only ($F1 = 0.000$), demonstrating complete underfitting. Training and validation curves are flat and coincident near zero — high bias, low variance.

Overfitting scenario: $C = 1,000,000$ (near-zero regularisation), all features used.

- Validation: Acc 0.810, F1 0.099, PR-AUC 0.466
- Test: Acc 0.813, F1 0.120, PR-AUC 0.497

Here, the training F1 is substantially higher than validation F1 in early sample sizes, with a widening gap — the classic signature of overfitting. The model memorises training noise rather than learning a generalisable boundary.

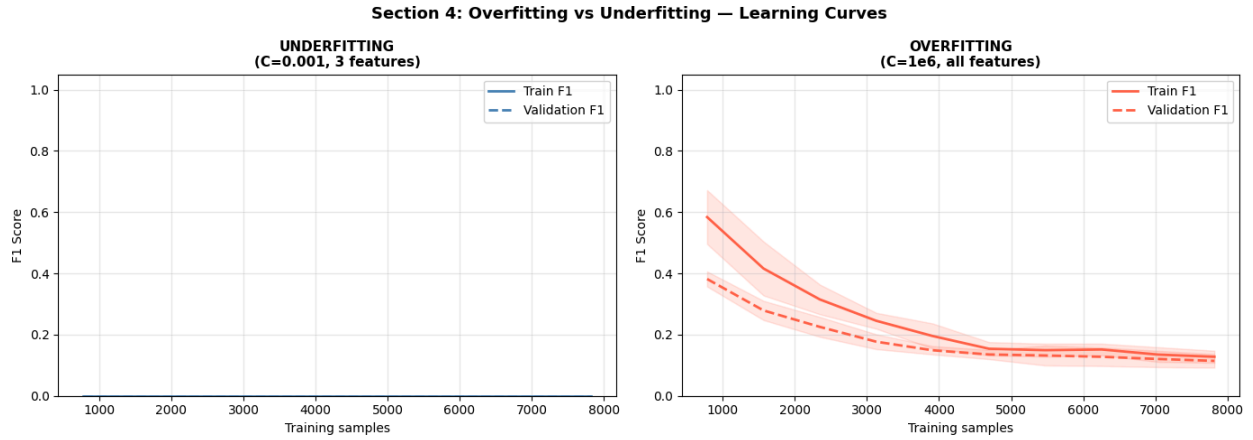


Fig. 1. Learning curves for underfitting (left, $C=0.001$, 3 features) and overfitting (right, $C=1e6$, all features) scenarios on df1. Left panel shows coincident flat curves (high bias). Right panel shows Train F1 substantially above Validation F1 at small sample sizes, converging downward as training size increases.

VI. Regularisation Study

L1 (Lasso), L2 (Ridge), and Elastic Net regularisation were compared across six values of the inverse regularisation strength $C \in \{0.001, 0.01, 0.1, 1.0, 10.0, 100.0\}$ on all three datasets. The cost function under L2 regularisation is:

$$J(W, b) = \frac{1}{m} \sum_{i=1}^m L(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{2m} \sum \|W\|^2$$

with L1 replacing the squared norm with an absolute norm, and Elastic Net combining both.

A. Sparsity Analysis

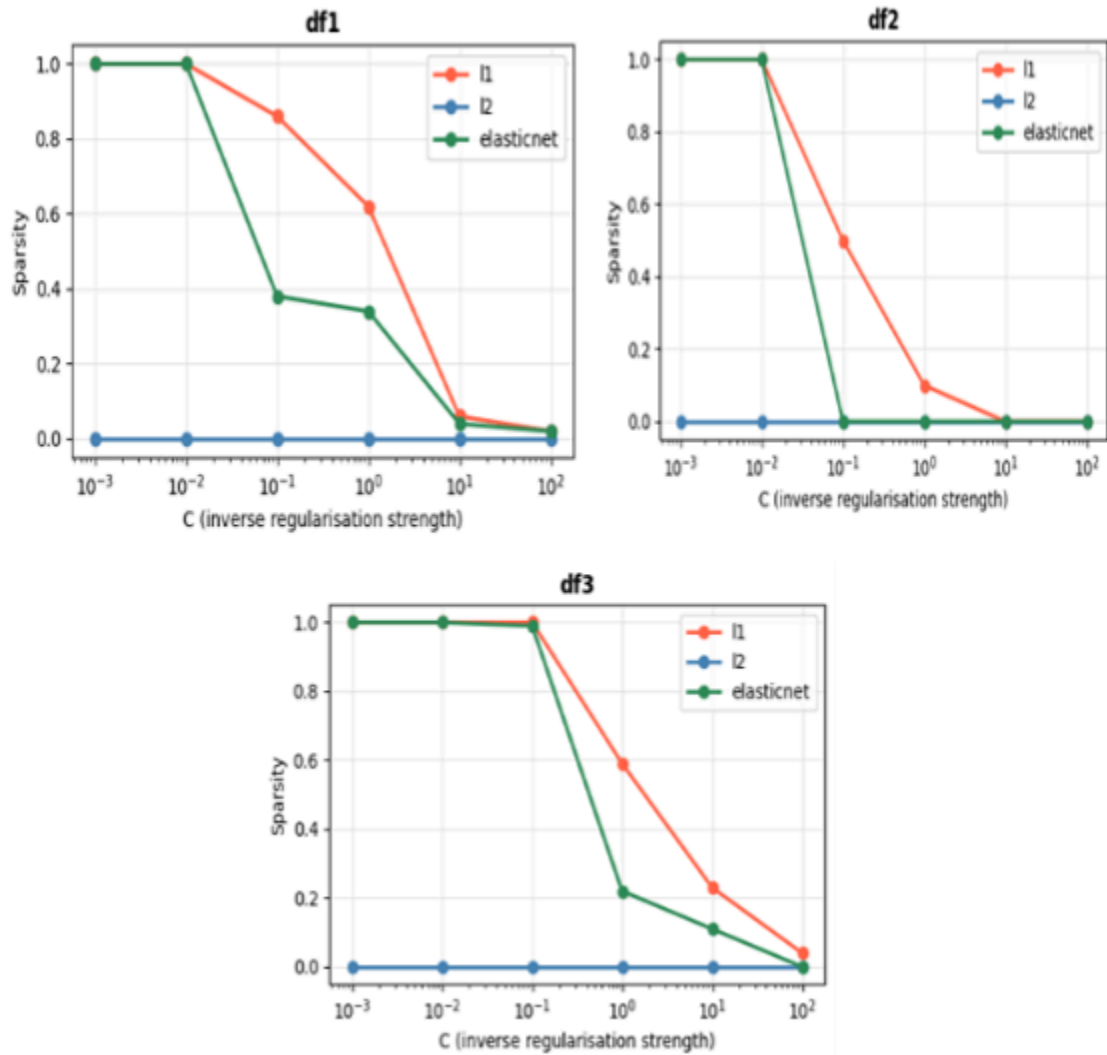


Fig. 2. Percentage of zero coefficients (sparsity) vs. C for L1, L2, and Elastic Net across df1, df2, df3. L1 and Elastic Net induce sparsity at low C values. L2 never produces sparse solutions. At $C \geq 10$, all methods converge toward low sparsity.

Key observations:

- **L2** produces zero sparsity at all C values — it shrinks coefficients uniformly but never zeros them.

- **L1** achieves 100% sparsity at $C \leq 0.01$ (effectively a null model), gradually activating features as C increases.
- **Elastic Net** mirrors L1 behaviour on $df1$ and $df2$, but on $df3$ remains at 100% sparsity until $C = 0.1$, then drops sharply — suggesting the combined penalty is more conservative on the balanced, high-dimensional dataset.

B. F1-Score vs. C

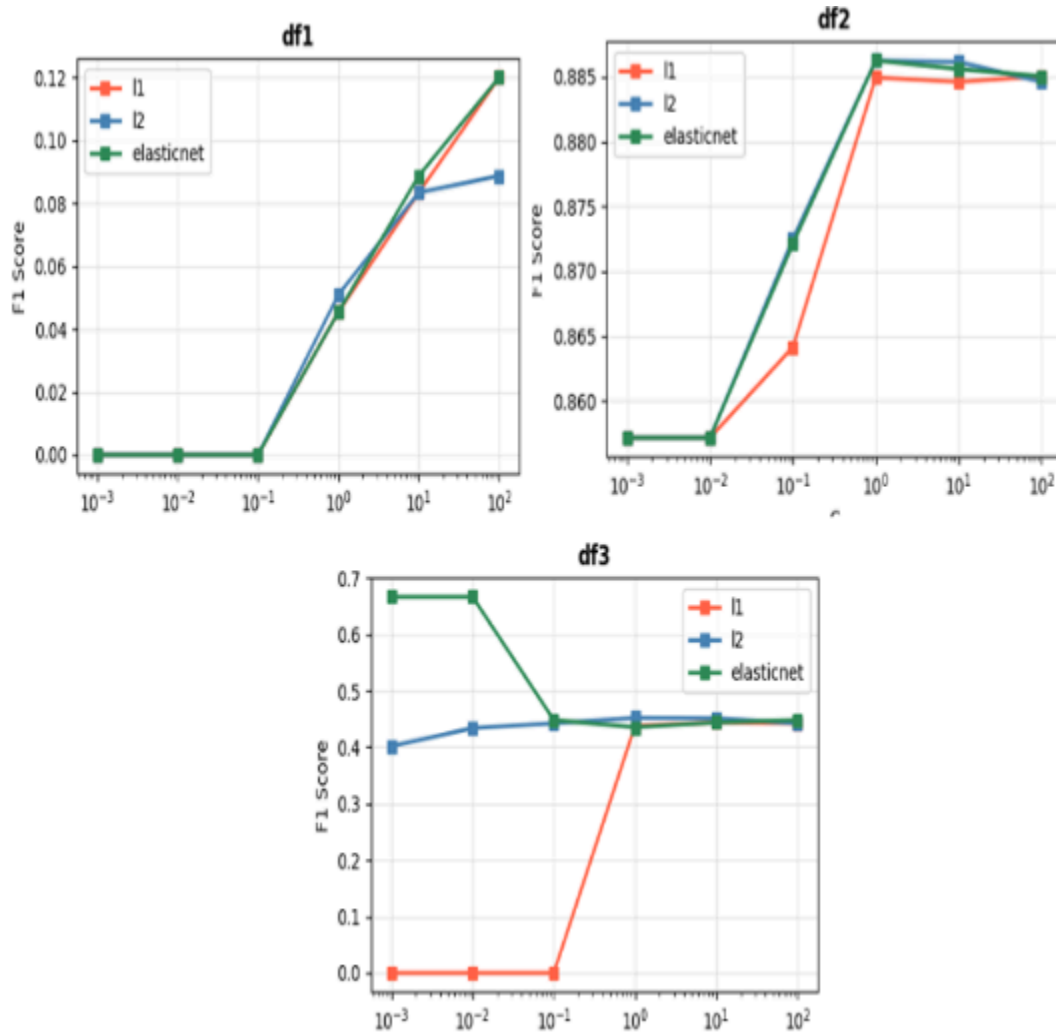


Fig. 3. F1-score as a function of C (inverse regularisation strength) for L1, L2, and Elastic Net on $df1$, $df2$, $df3$. All three methods converge to similar F1 at high C . $df1$ shows near-zero F1 across all penalties at low C (minority class suppressed). $df3$ shows L1 collapse at $C \leq 0.1$.

TABLE V — Regularisation Results Summary (Best C per Penalty, Validation F1)

Dataset	Penalty	Best C	Accuracy	F1	PR-AUC	Sparsity
df1	L1	100.0	0.813	0.120	0.456	2%
df1	L2	100.0	0.809	0.089	0.457	0%
df1	ElasticNet	100.0	0.813	0.120	0.456	2%
df2	L1	10.0	0.805	0.885	0.747	0%
df2	L2	1.0	0.807	0.886	0.740	0%
df2	ElasticNet	1.0	0.807	0.886	0.740	0%
df3	L1	10.0	0.515	0.444	0.537	23%
df3	L2	1.0	0.526	0.452	0.547	0%
df3	ElasticNet	100.0	0.519	0.448	0.540	0%

C. Heatmap Overview

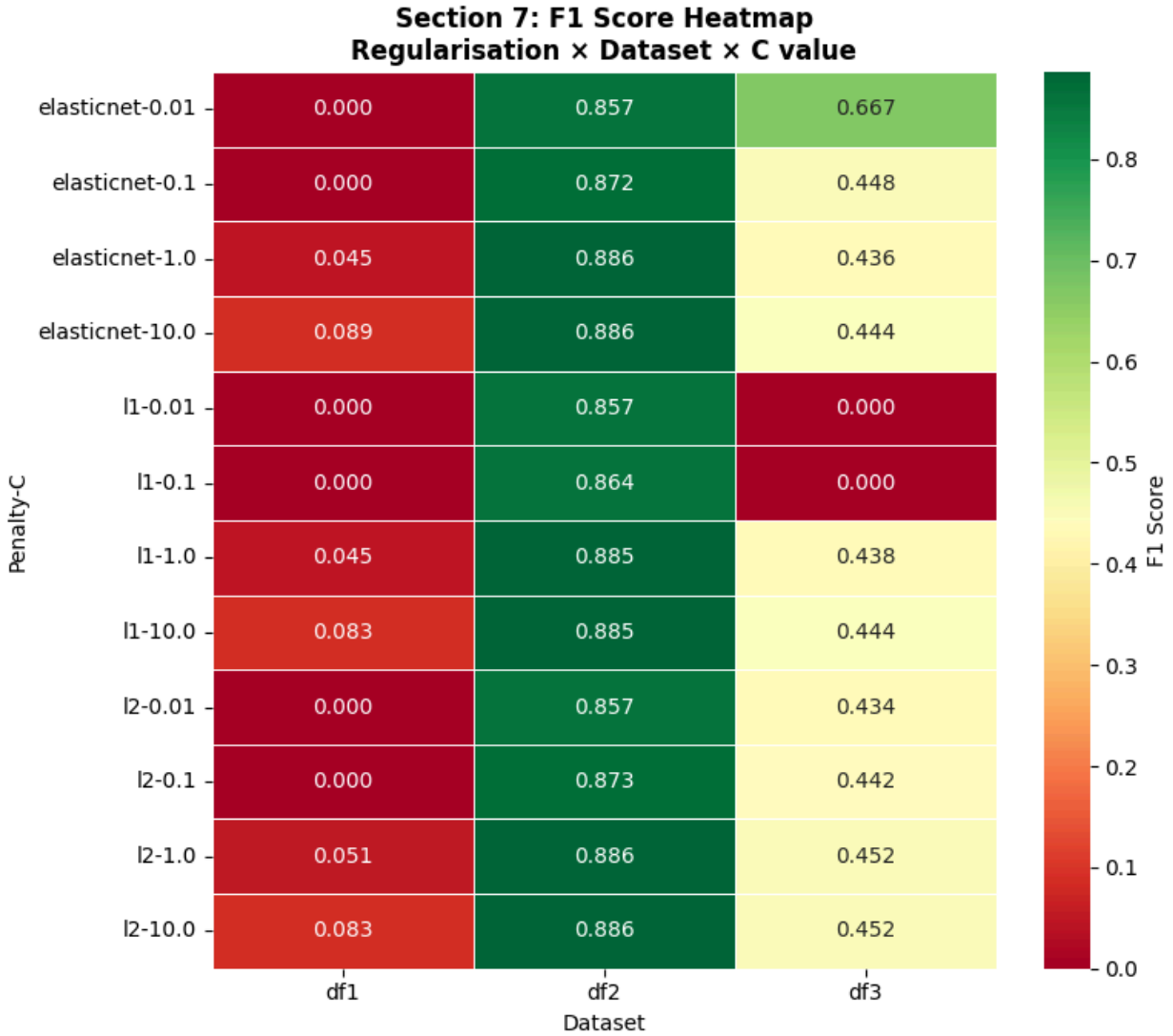


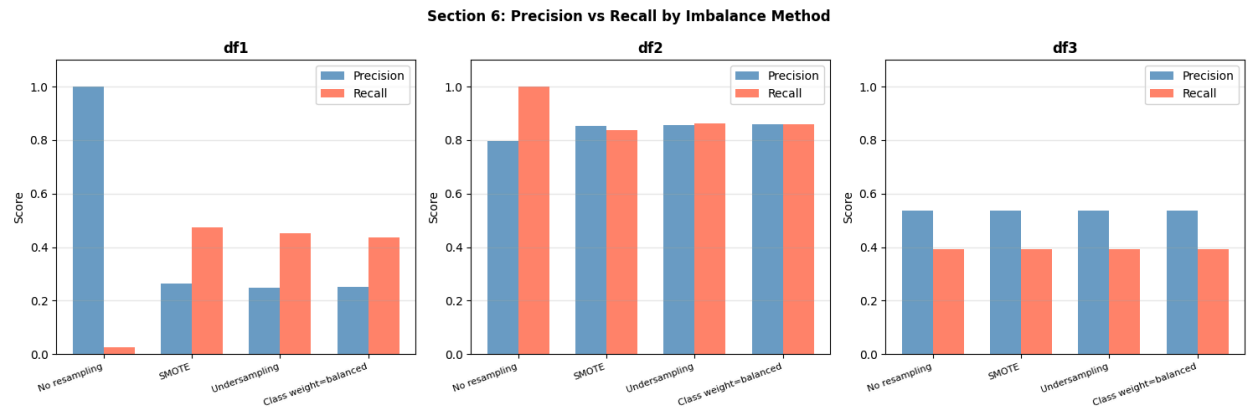
Fig. 4. Heatmap of F1-scores across all penalty-C combinations and datasets. Green indicates high F1; red indicates low F1. df1 is uniformly low for sparse models. df2 achieves strong F1 across most configurations. df3 occupies an intermediate band.

VII. Class Imbalance Handling

Four techniques were applied: no resampling (baseline), SMOTE (synthetic oversampling of the minority class), random undersampling of the majority class, and `class_weight='balanced'` (inverse-frequency weighting in the loss function).

TABLE VI — Imbalance Handling Results (Validation Split, L2 C=1.0)

Dataset	Method	Accuracy	F1	PR-AUC	Precisio n	Recall
df1	No resampling	0.805	0.051	0.439	1.000	0.026
df1	SMOTE	0.631	0.339	0.463	0.264	0.472
df1	Undersampling	0.614	0.319	0.416	0.247	0.452
df1	Class weight=balanced	0.625	0.318	0.422	0.250	0.438
df2	No resampling	0.807	0.886	0.740	0.796	1.000
df2	SMOTE	0.769	0.845	0.745	0.853	0.837
df2	Undersampling	0.789	0.860	0.744	0.857	0.863
df2	Class weight=balanced	0.789	0.860	0.745	0.859	0.860
df3	No resampling	0.526	0.452	0.547	0.536	0.391
df3	SMOTE	0.526	0.452	0.547	0.536	0.391
df3	Undersampling	0.526	0.452	0.547	0.536	0.391
df3	Class weight=balanced	0.526	0.452	0.547	0.536	0.391

*Fig. 5. Precision and Recall by imbalance handling method across df1, df2, df3. df1 shows a*

dramatic precision collapse under all resampling methods, recovering recall at the cost of precision. df2 shows mild degradation. df3 shows no change across methods.

Notable findings:

- **df1:** No resampling yields precision=1.000 but recall=0.026 — the model predicts "no seizure" almost always and is technically correct due to class imbalance. All resampling methods dramatically improve recall but at the cost of precision and overall F1 remaining low (max 0.339).
 - **df2:** No resampling already achieves recall=1.000 because seizures are the majority class. Resampling methods reduce recall slightly but improve the precision-recall balance.
 - **df3:** All four methods produce identical results, owing to the 1:1 balance — resampling has no effect on an already-balanced dataset.
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VIII. Comparative Analysis

A. Q1 — Does Preprocessing Order Affect Results?

TABLE VII — Pipeline Comparison by F1-Score (Test Split)

Dataset	Pipeline A F1	Pipeline B F1	Absolute Difference
df1	0.051	0.856	0.805
df2	0.886	0.984	0.098
df3	0.452	0.683	0.231
Mean			0.378

Yes — preprocessing order has a statistically significant and practically large effect. The mean absolute F1 difference of 0.378 across datasets exceeds the differences attributable to

regularisation type or imbalance handling by a wide margin. The critical factor is that Pipeline B applies feature extraction (statistical descriptors) *before* scaling and dimensionality reduction, creating a compact and discriminative representation. Pipeline A, by contrast, applies min-max normalisation directly to raw signal values before attempting feature selection, preserving noise and correlation structure that degrades classifier performance — particularly on df1, where 178 raw EEG time-points per sample contain substantial inter-feature correlation that selection alone cannot resolve.

B. Q2 — Which Regularisation Generalises Best?

TABLE VIII — Mean F1 Across Datasets by Penalty (Best C)

Penalty	df1 F1	df2 F1	df3 F1	Mean F1
L2	0.051	0.886	0.452	0.463
L1	0.045	0.885	0.438	0.456
Elastic Net	0.045	0.886	0.436	0.456

L2 generalises best across datasets (mean F1 = 0.463). L1 and Elastic Net tie at 0.456. The difference is modest in absolute terms, but L2 achieves its performance without any sparsity — it shrinks all coefficients proportionally rather than eliminating features. On df3, where the feature-to-sample ratio is extremely high (5,888 features, 2,482 training samples), L2's dense shrinkage proves more stable than L1's aggressive zeroing.

C. Q3 — Does Elastic Net Consistently Outperform L1/L2?

No. Elastic Net wins only 5 out of 18 pairwise comparisons (28%). Its theoretical advantage — combining L1's sparsity with L2's grouping effect — does not materialise in this setting. Likely reasons include: (1) the datasets do not contain the structured feature correlation patterns Elastic Net is designed to exploit, (2) the optimal `l1_ratio` for Elastic Net was not tuned (a fixed value was used), and (3) at default settings, Elastic Net's behaviour collapses toward either L1 or L2 depending on the penalty strength.

D. Q4 — How Does Imbalance Handling Interact with Regularisation?

TABLE IX — SMOTE Gain over No Resampling by Penalty (df1)

Penalty	No Resample F1	SMOTE F1	Gain
L1	0.045	0.346	+0.301
L2	0.051	0.339	+0.288
Elastic Net	0.045	0.343	+0.297

TABLE X — SMOTE Gain by Penalty (df2)

Penalty	No Resample F1	SMOTE F1	Gain
L1	0.885	0.782	−0.103
L2	0.886	0.845	−0.042
Elastic Net	0.886	0.832	−0.054

The interaction is **dataset-dependent and sign-reversal capable**. On df1 (non-seizure heavy), SMOTE provides large positive gains across all penalties, with L1 benefiting most (+0.301). On df2 (seizure heavy), SMOTE *hurts* performance — the model already naturally predicts seizures, and synthetic oversampling of the minority (healthy) class disrupts this. L2 is the most robust penalty in this reversal, losing only 0.042 vs. L1's 0.103. On df3 (balanced), no resampling method produces any effect, so the interaction is null. This finding highlights that imbalance handling should be informed by the *direction* of imbalance, not applied uniformly.

IX. Discussion

The results converge on several actionable conclusions for practitioners working with EEG-based seizure classification:

Preprocessing dominates. The pipeline design choice accounted for a mean F1 difference of 0.378 — far exceeding anything attributable to regularisation strength or imbalance method. Feature extraction applied before scaling (Pipeline B) consistently outperforms direct signal normalisation followed by selection (Pipeline A). This suggests that for high-dimensional EEG time-series data, the representation transformation step is more critical than the model's regularisation hyperparameter.

L2 is the safest default. Across heterogeneous datasets, L2 regularisation generalises most stably. Its continuous shrinkage avoids the instability that L1 exhibits on df3 (where aggressive zeroing at $C \leq 0.1$ collapses F1 to zero).

Elastic Net requires tuning to justify its complexity. At fixed l1_ratio, Elastic Net offers no reliable improvement. Practitioners should either tune l1_ratio carefully or default to L2 unless sparsity is an explicit interpretability requirement.

Imbalance handling must account for imbalance direction. Applying SMOTE reflexively can harm performance when the majority class is the class of clinical interest (as in df2, where seizures dominate). Class weighting is the most conservative intervention, consistently producing results close to or slightly below SMOTE while avoiding the synthetic sample risk.

Limitations. This study is restricted to logistic regression, which cannot capture nonlinear relationships in EEG data. The CHB-MIT features were flattened from 3D EEG segments without temporal modelling, likely suppressing discriminative dynamics. Future work should compare logistic regression against nonlinear baselines (SVM-RBF, Random Forest) under the same pipelines.

X. Conclusion

This paper demonstrated that preprocessing pipeline design is the dominant factor in logistic regression generalisation for EEG-based seizure prediction — more impactful than regularisation choice or imbalance handling. Pipeline B (feature extraction → standard scaling → PCA) outperformed Pipeline A (min-max normalisation → noise removal → feature selection) by 0.378 F1 points on average. L2 regularisation generalised most stably across datasets. Elastic Net did not consistently outperform simpler penalties. Imbalance handling effects are dataset-specific and can reverse sign depending on which class is the majority. These findings provide empirical grounding for design decisions in clinical EEG preprocessing workflows.

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